Department of Engineering Science
School of Science & Technology



CES 552 Network Architecture
Introduction

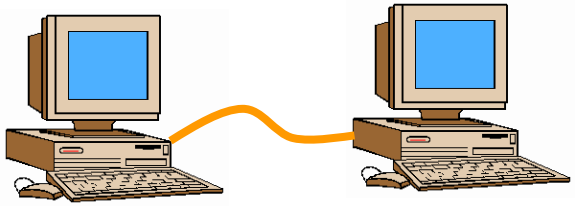
Dr. Jingxian Wu
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OUTLINE

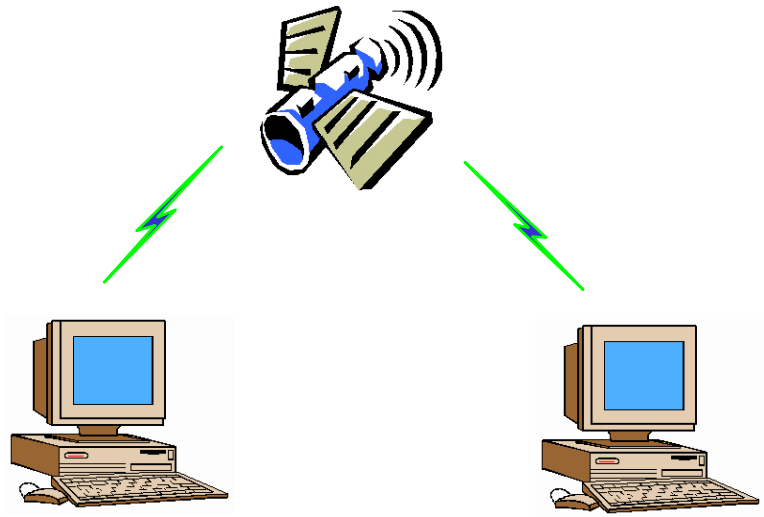
- **Network Concepts**
- **Physical Layer**
- **Data Link Layer**
- **Media Access Control Sublayer**
- **Network Layer**

CONCEPTS: WHAT IS COMPUTER NETWORK?

- **Computer Network: a collection of autonomous computers interconnected by certain technologies.**
  - Autonomous: independent, self-sufficient, self-ruling.
  - Interconnection: capable of exchanging information with one another.
    - Via copper wire, fiber optics, microwaves, infrared, etc.
- **Examples:**



Network 1



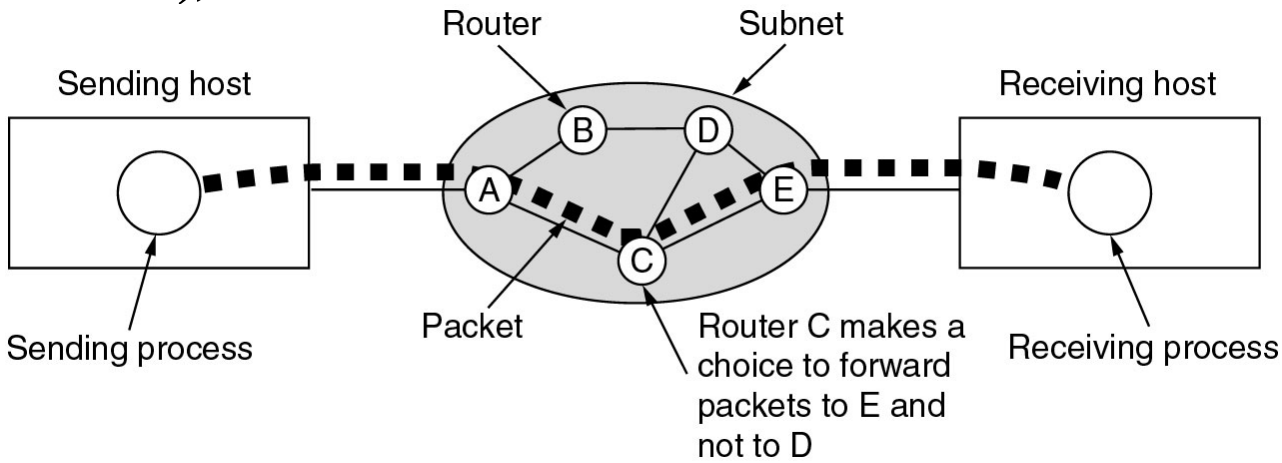
Network 2

CONCEPTS: CLASSIFICATION

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

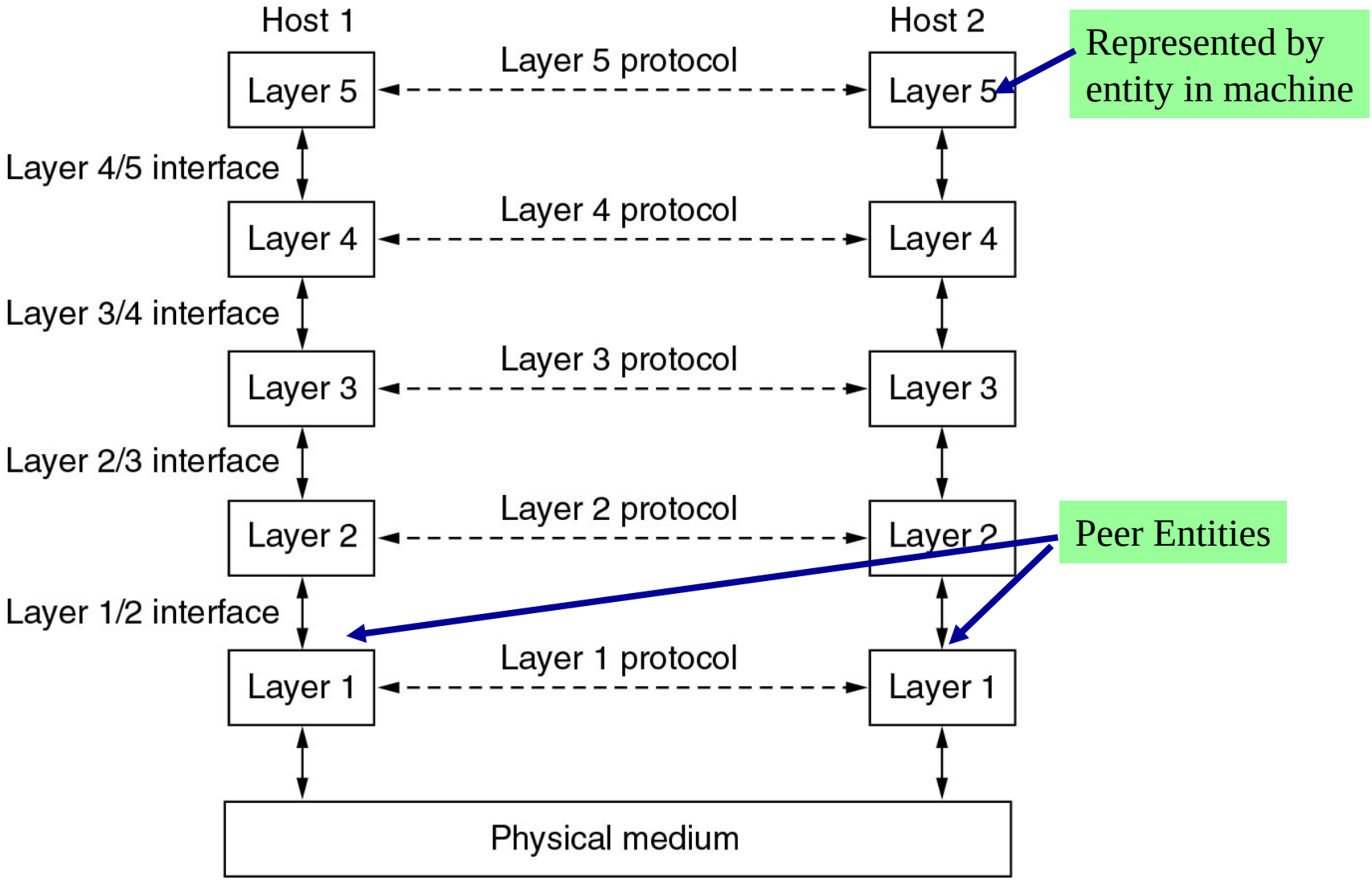
CONCEPTS: CLASSIFICATION

- **PAN**
  - Examples: bluetooth, IEEE 802.15
- **LAN**
  - Ethernet (IEEE 802.3), wireless LAN (IEEE 802.11)
- **MAN**
  - WiMax (IEEE 802.16), Cable TV
- **WAN**



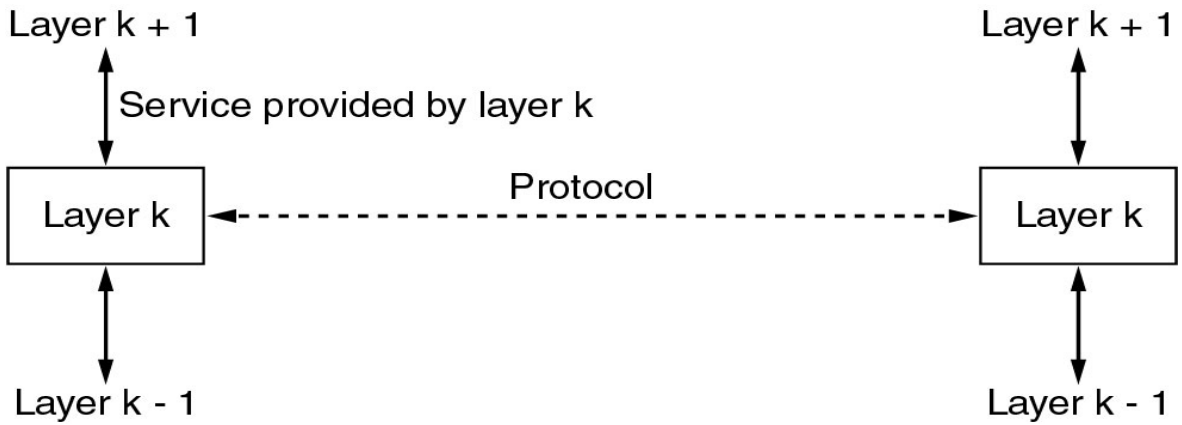
- **internetwork**
  - Internet

CONCEPTS: PROTOCOL HIERARCHIES



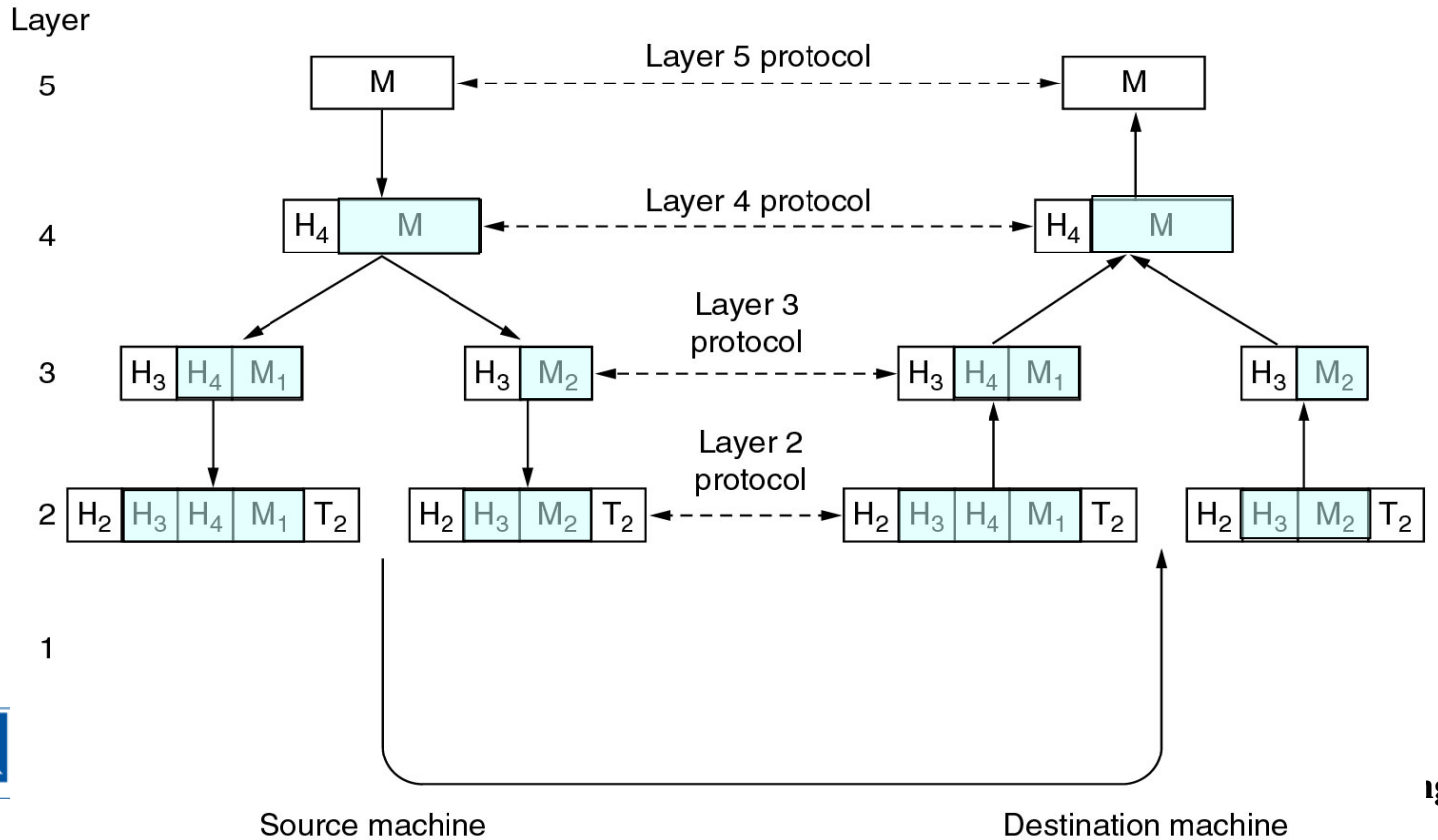
CONCEPTS: SERVICES V.S. PROTOCOL

- **Service: A set of primitives (operations) that a layer provides to the layer above it.**
  - Vertical: Interface between two layers.
  - Visible to user (upper layer)
- **Protocol: A set of rules governing the format and handling of packets exchanged by peer entities.**
  - Horizontal: interface between peer entities.
  - Invisible to user (upper layer)



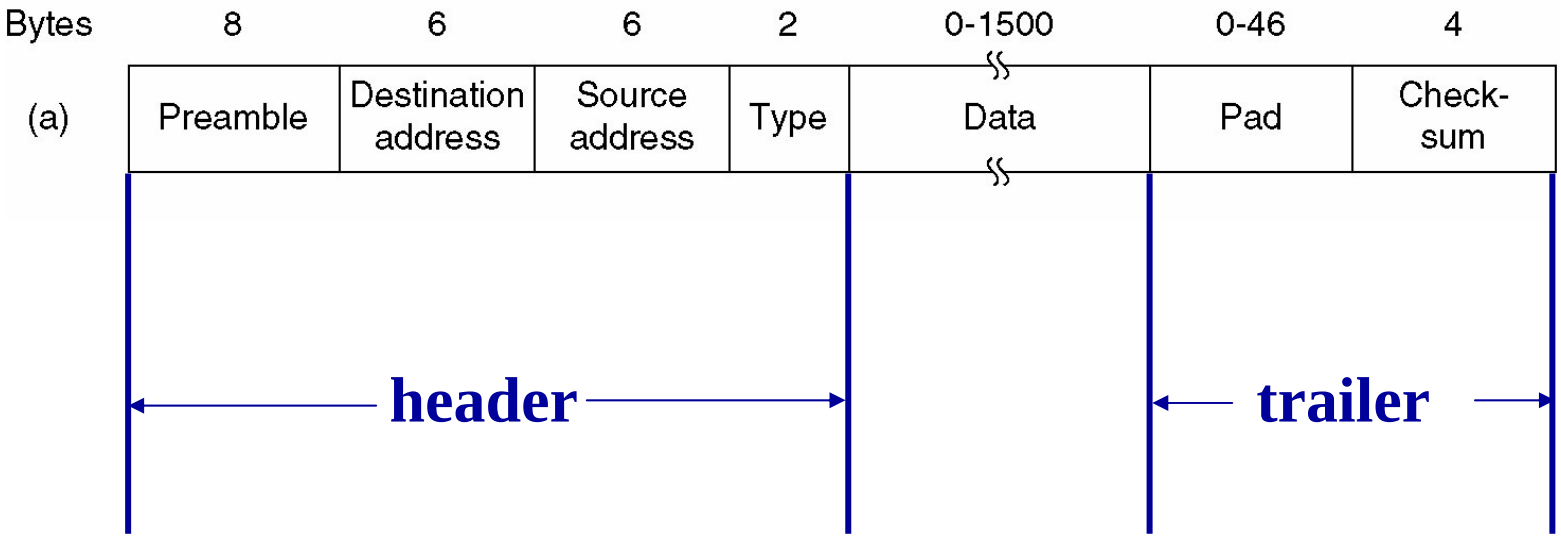
CONCEPTS: PROTOCOL DATA UNIT

- **PDU: a unit of data that is specified by a protocol of a given layer and delivered among peer entities**
- **PDU comprises**
  - Control information: How peer entity should handle PDU.
  - Data: Upper layer PDU.



CONCEPTS: PDU

- **Example: Ethernet PDU in MAC sublayer**

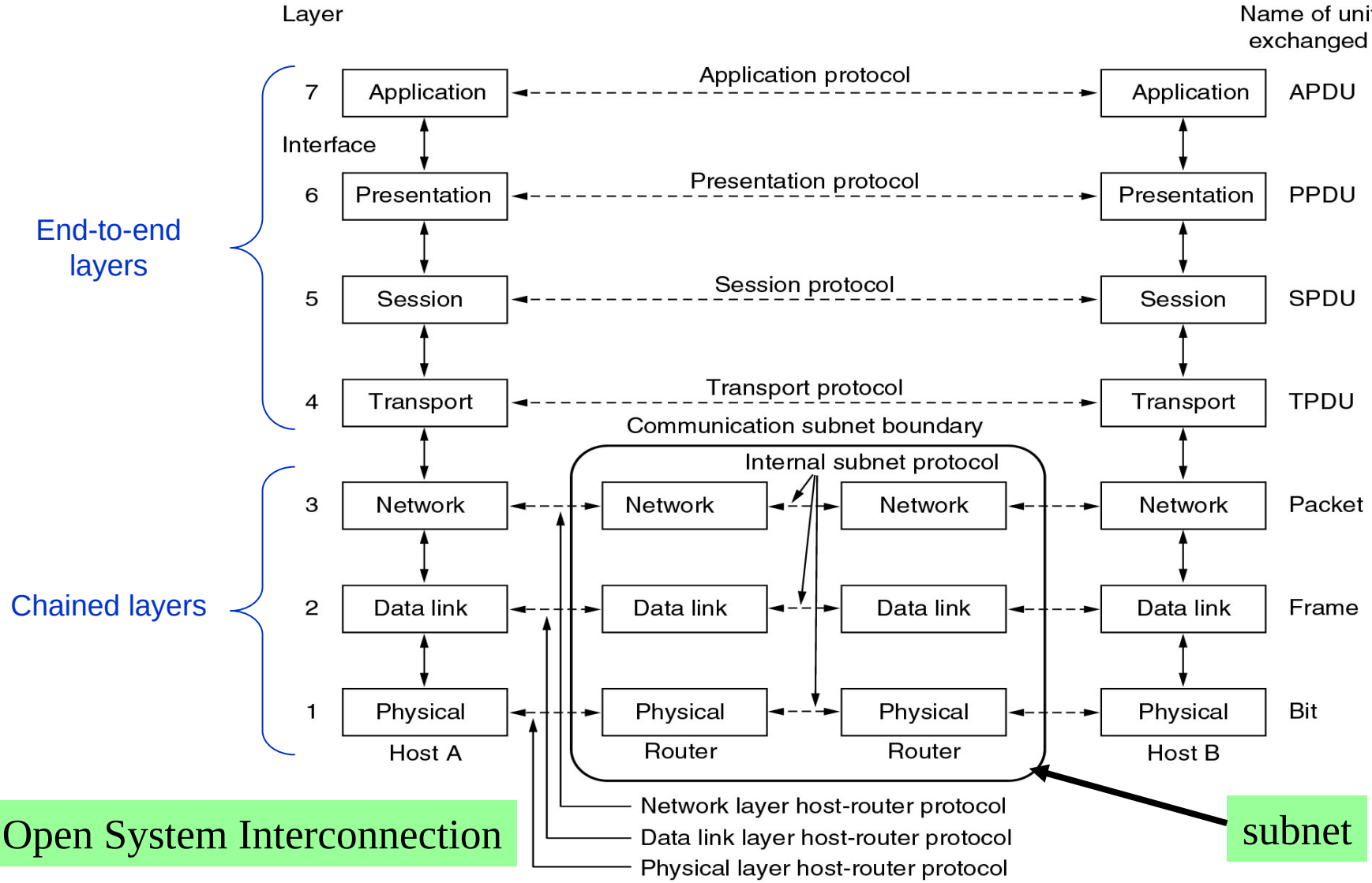


CONCEPTS: SERVICES

- **Connection Oriented Service (3 steps)**
  - 1. Establish a connection before transmission
  - 2. Transmit data through the connection
  - 3. Release the connection after the transmission is done
  - E.g. telephone system, TCP (Transport Control Protocol)

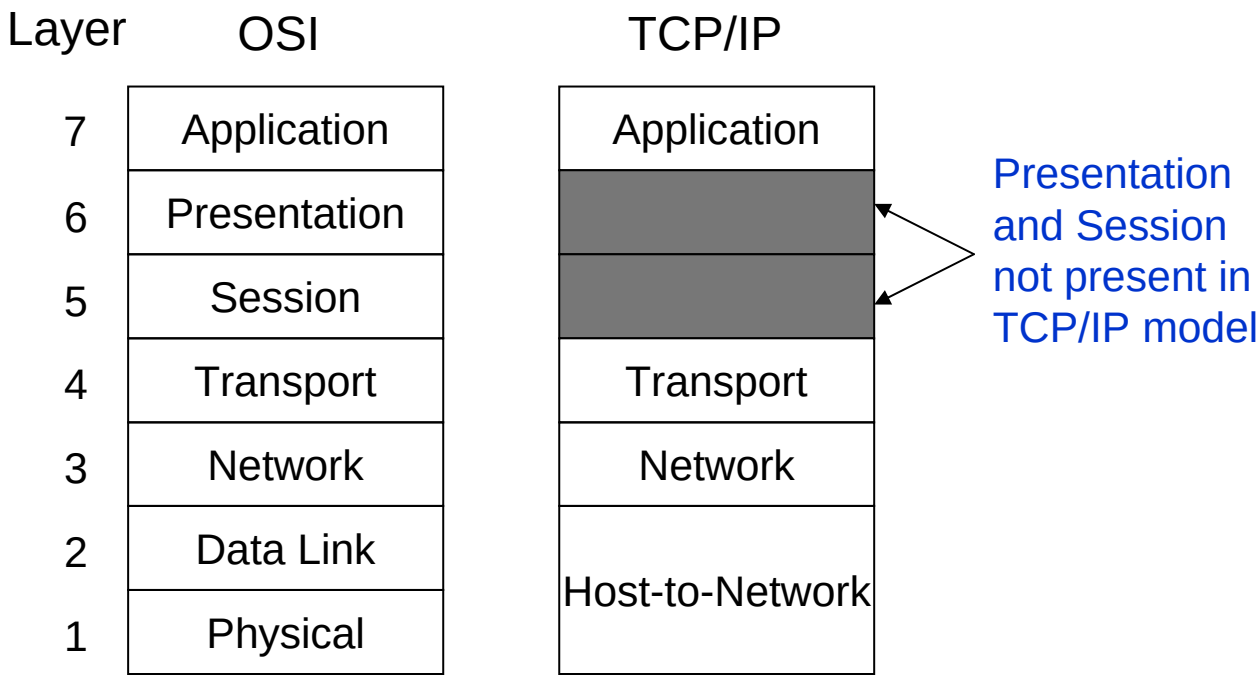
- **Connectionless Service (Datagram service)**
  - Send out packet without connection
  - Each packet is individually routed
  - E.g. IP (Internet Protocol), UDP (User Datagram Protocol)

CONCEPTS: OSI REFERENCE MODEL

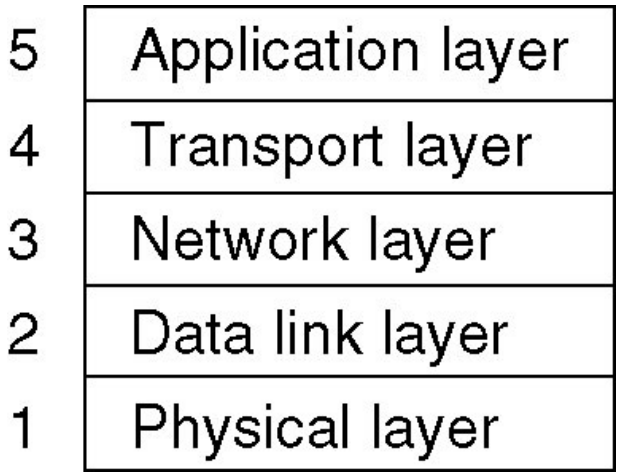


Open System Interconnection

CONCEPTS: TCP/IP MODEL

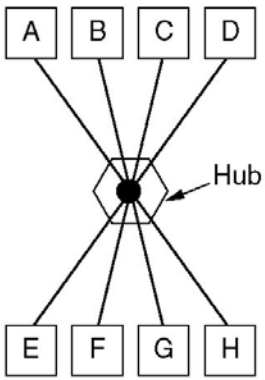


CONCEPTS: HYBRID MODEL

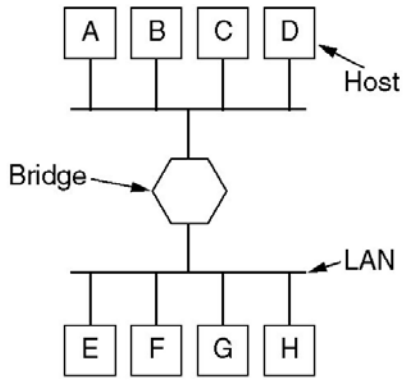


CONCEPTS: TERMINOLOGIES

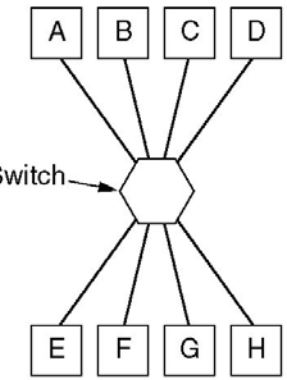
- **Terminologies: Repeater, Hub, Bridge, Switch, Router, Gateway**
  - Repeater: Analog device amplifying signal (Physical Layer)
    - Doesn't care about frame contents: only cares voltage
  - Hub: Connecting multiple hosts like a shared cable (Physical Layer)
    - Doesn't care about frame contents
  - Bridge: Connecting two or more LANs (Data Link Layer)
    - Examine the frame headers, route based on frame address (e.g. MAC address)
    - Different LANs can have different data link layer
  - Switch: Connecting two or more hosts (Data Link Layer)
    - Examine the frame headers, route based on frame address



(a)



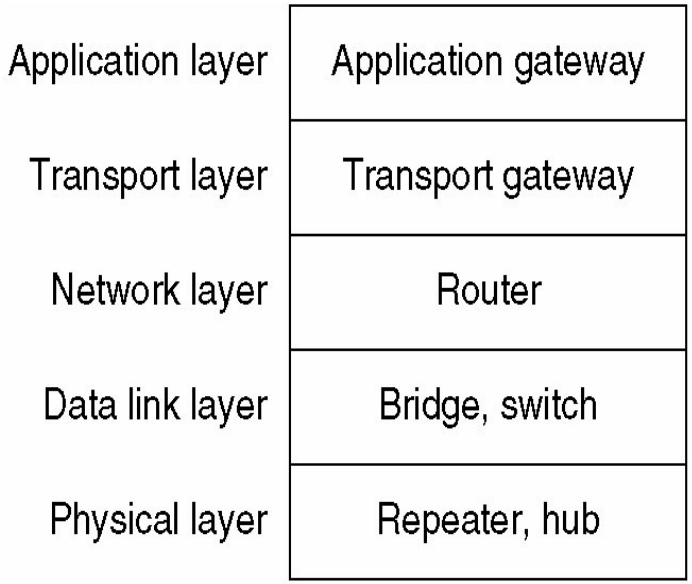
(b)



(c)

CONCEPTS: TERMINOLOGIES

- **Terminologies (Cont'd)**
  - Router: Routing packets based on network layer header
    - E.g. routing based on IP address.
    - Multi-protocol router is also called gateway
  - Transport Gateway: Connecting two hosts with different transport layer protocols
    - E.g. One host using TCP/IP protocol talk with another one using ATM transport protocol.
  - Application Gateway: Translating data from one format to another.
    - E.g. E-mail gateway: translate regular email message into SMS message for mobile phones.



CONCEPTS: STANDARDS

- **ITU: International Telecommunication Union**
  - Telecommunications, V.92 (modem), V.24 (RS-232), X.25, SS7, …
- **ISO: International Standards Organization**
  - Computer and other areas, OSI
- **IAB: Internet Architecture Board (IETF and IRTF)**
  - Internet, RFC (Request for Comments, TCP/IP),
- **IEEE: Institute of Electrical and Electronics Engineers**
  - Networking and other electrical areas, IEEE 802.3, IEEE 802.11, IEEE 802.16

OUTLINE

- Network Concepts
- **Physical Layer**
- Data Link Layer
- Media Access Control Sublayer
- Network Layer

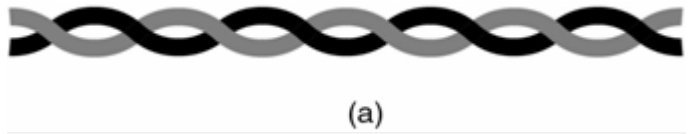
PYSICAL LAYER

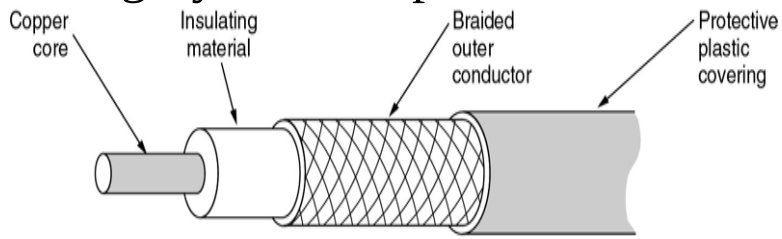
- **Functions:**
  - Transmit information over the physical medium
    - Make sure that a ‘1’ is sent out at transmitter, a ‘1’ is received at receiver.
  - Implementations depend on the properties of the medium
  - Mechanical Interface: # of pins, shape of connector, etc.
  - Electrical Interface: volts used to represent ‘1’, etc.

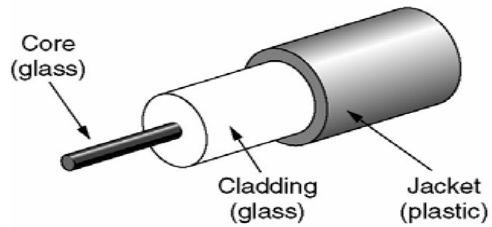
- **Example protocols**
  - RS-232 (serial port)
  - V.92 (dial-up modem)
  - USB (Universal Serial Bus)

PHYSICAL LAYER: TRANSMISSION MEDIA

- **Guided media**
  - Removable media
    - DVD, flash drive
  - Twisted Pair
    - Phone line, Ethernet cable
    - Category 3: BW up to 16MHz; Category 5: BW up to 100MHz
  - Coaxial Cable
    - TV distribution
    - BW close to 1GHz
  - Fiber Optic Cable
    - Network backbone

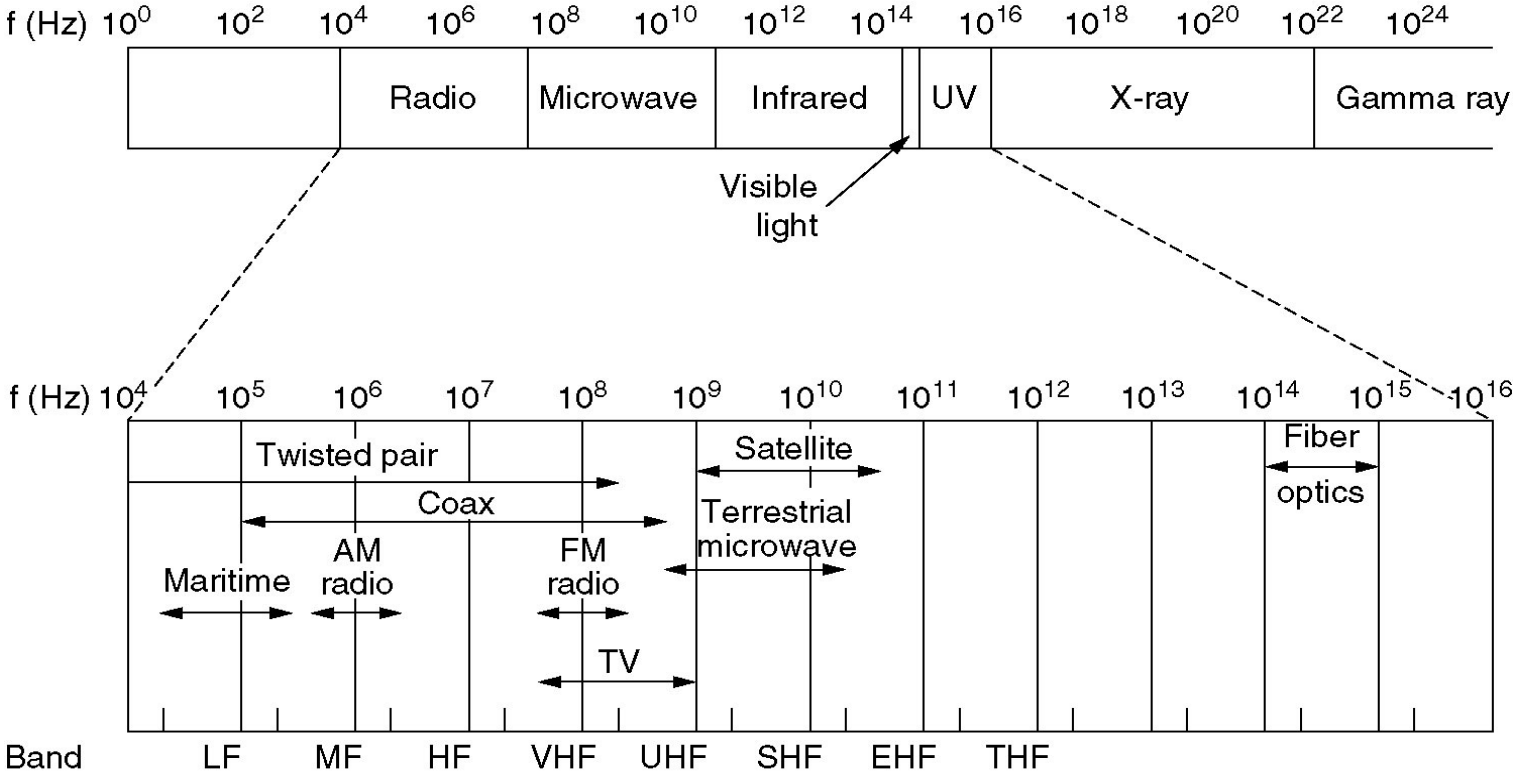






PHYSICAL LAYER: TRANSMISSION MEDIA

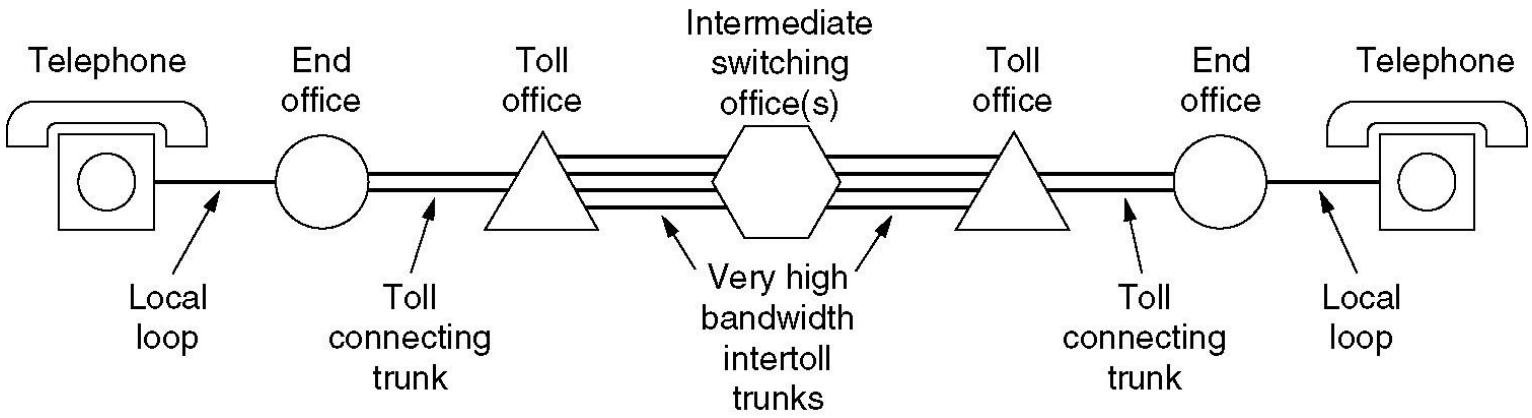
- **Wireless: the propagation of electromagnetic waves**



Industrial Scientific Medical (ISM) band: 900MHz, 2.4GHz, 5.8GHz

PHYSICAL LAYER: PSTN EXAMPLE

- **Public Switched Telephone Network (PSTN)**
  - Local loop (dial-up, ADSL)
  - Switching office
  - Trunk (fiber optic)
    - Also used as backbone for Internet



OUTLINE

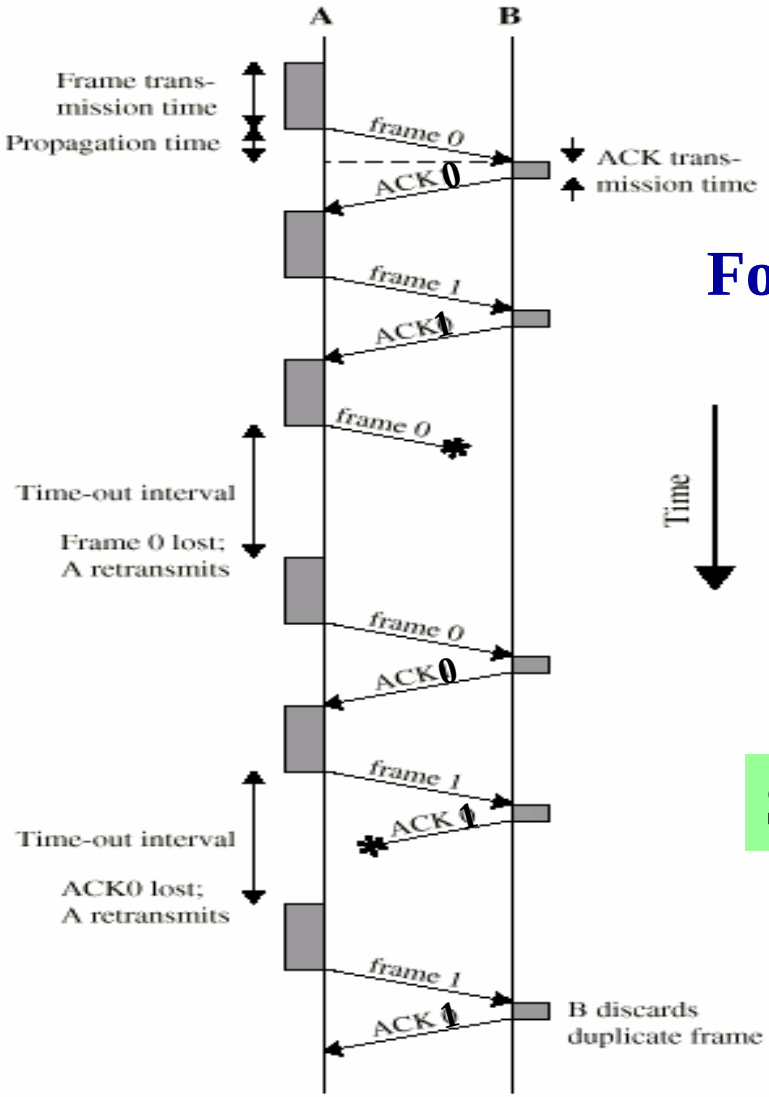
- Network Concepts
- Physical Layer
- **Data Link Layer**
- Media Access Control Sublayer
- Network Layer
- Example Networks

DATA LINK LAYER

- **Functions:**
  - Provide connection-oriented or connectionless service to upper layer
  - Framing:
    - encapsulate bit streams from physical layer into frames
  - Flow control:
    - keep fast transmitter from swamping slow receiver (link level)
    - Different from congestion control (avoid congestion in the whole subnet, system level)
  - Error control
    - Error detection (parity check)
    - Error correction (linear block code, cyclic redundancy check)
    - Acknowledgement, retransmission

Deals with communication between two adjacent machines.

DATA LINK LAYER: STOP AND WAIT



For both error control and flow control.

Sliding window with window size 1

DATA LINK LAYER: EXAMPLE PROTOCOLS

- **HDLC (High-level data link control)**
  - A connection oriented data link protocol
  - Using sliding window for error control and flow control
  - Still in use (X.25, used to provide services to private leased lines from telephone company)
- **PPP (Point-to-point protocol)**
  - Used in dial-up Internet connection
  - Set up a point-to-point link between the dial-up modem and the dial-up server in PSTN switching offices.

OUTLINE

- Network Concepts
- Physical Layer
- Data Link Layer
- **Media Access Control Sublayer**
- Network Layer

MAC

- **Functions**
  - Determine how to allocate a single broadcast channel among multiple competing users.

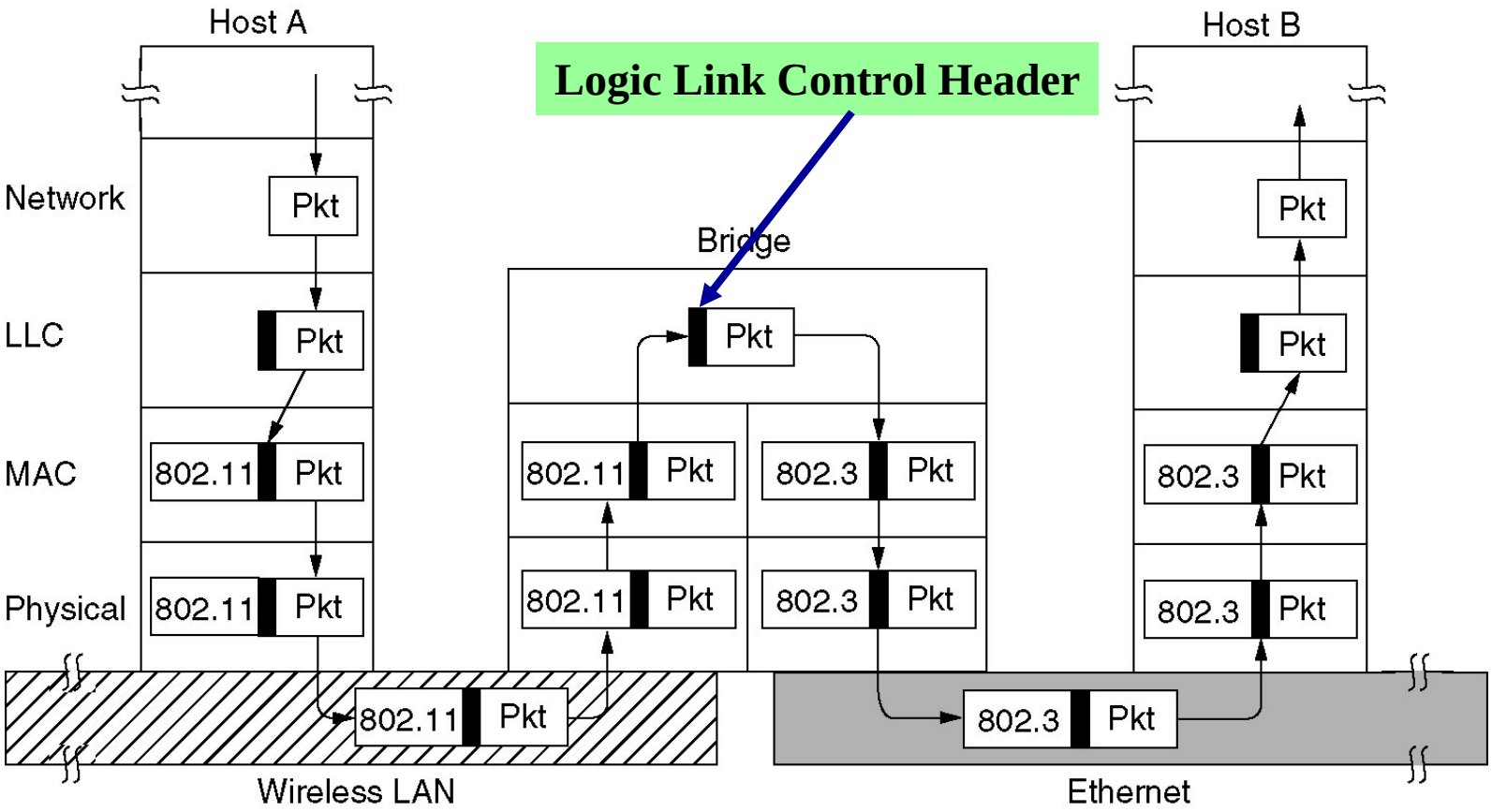
- **Broadcast link v.s. Point-to-point link**
  - Broadcast link: a single communication channel shared by all the machines
  - Point-to-point link: a channel connecting individual pair of computers.

- **Most LANs use broadcast link**
  - Ethernet, Wireless LAN, etc.
  - All hosts use one shared media

MAC: PROTOCOLS

- **ALOHA**
  - Transmit whenever there are packets
  - Two or more transmit at the same time ➔ collision
  - If collision happens, wait for a random time and retransmit
- **Collision Sensing Multiple Access (CSMA)**
  - Listen to channel before transmission
  - If there is transmission, host will backoff (Collision Sensing)
- **CSMA/CD (collision detection)**
  - Other than CS, sender senses the channel during transmission to detect collision (CD)
  - Used by Ethernet

MAC: BRIDGE


Host A
Host B
Logic Link Control Header
Network
LLC
MAC
Physical
Bridge
Pkt
802.11
802.3
Wireless LAN
Ethernet

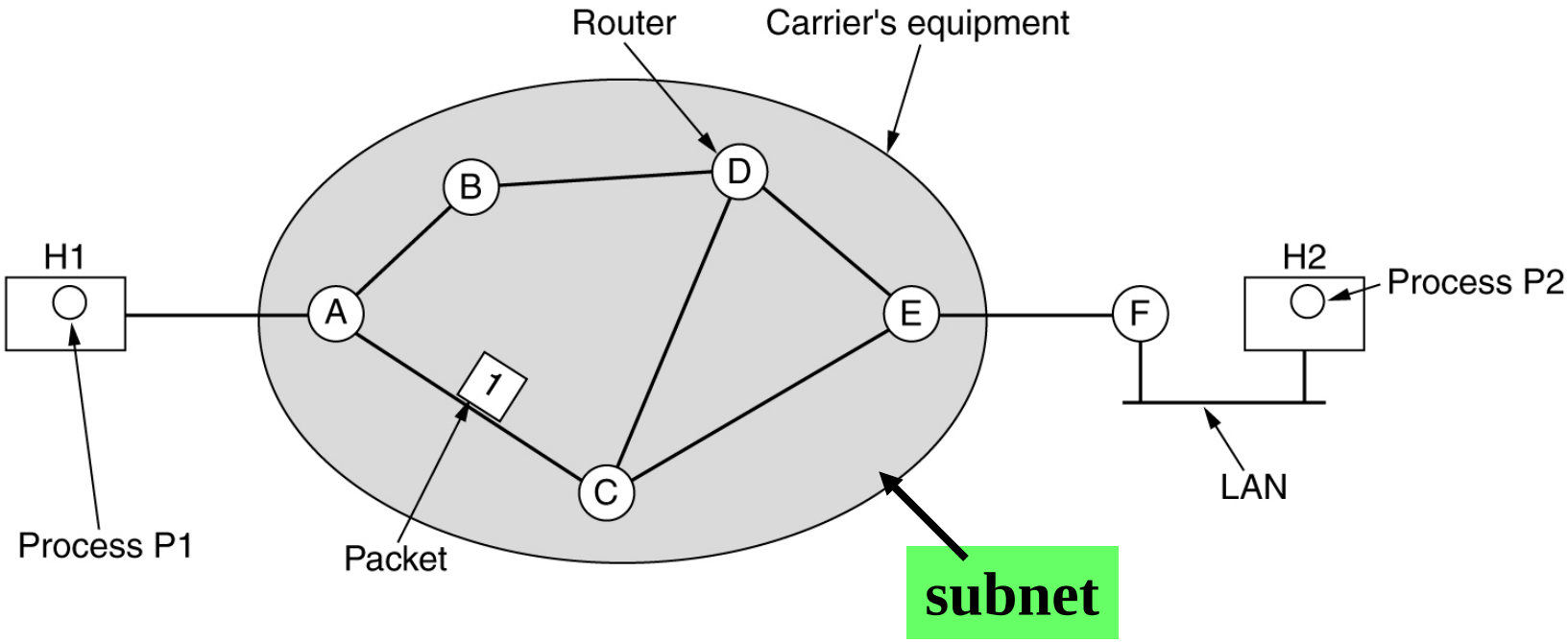
OUTLINE

- Network Concepts
- Physical Layer
- Data Link Layer
- Media Access Control Sublayer
- **Network Layer**

NETWORK LAYER

- **Functions**
  - Routing
    - Deliver the packet from source all the way to destination.
    - Find the appropriate path
  - Congestion control
    - Avoid overloading some of the lines in subnet
    - (flow control: avoid overloading the computer at the other end of the line)
  - Quality of Service
    - Guarantee the services provided to the end user
  - Internetworking
    - Connect different types of networks
  - Service to transport layer
    - Connectionless service
    - Connection oriented service (virtual circuit service)
    - Transport layer should be shielded from the number, type, and topology of the routers present

NETWORK LAYER



Routers can be in either subnet (owned by operator) or LAN (owned by end users).

NETWORK LAYER

- **Connectionless Service**
  - Datagram subnet
  - Store-and-forward
    - Packets received by router in its entirety.
    - Packets stored in router, checksum examined by router
    - Packet is forwarded to the next router
  - Flexible, reliable.
- **Connection Oriented**
  - Connection in network layer also called virtual circuit (VC)
    - Only true for network layer.
    - Generally, connection Oriented $\neq$ Virtual circuit
  - All the packets belonging to the same VC will follow the same route.
  - Easy for QoS control, congestion control.

- **Quality of Service**
  - Reliability
    - Low bit error rate
  - Delay
    - Low transmission delay
  - Jitter
    - Packets arriving at uniform intervals
  - Bandwidth
    - Large bandwidth ➔ High data rate.

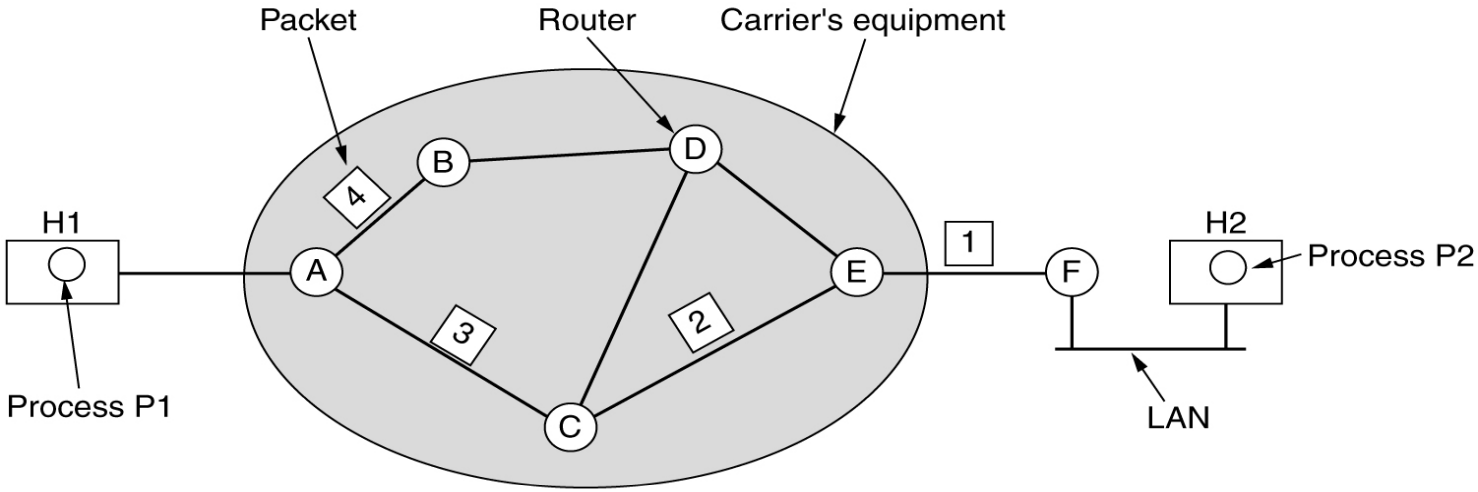
NETWORK LAYER

Application	Reliability	Delay	Jitter	Bandwidth
E-mail	High	Low	Low	Low
File transfer	High	Low	Low	Medium
Web access	High	Medium	Low	Medium
Remote login	High	Medium	Medium	Low
Audio on demand	Low	Low	High	Medium
Video on demand	Low	Low	High	High
Telephony	Low	High	High	Low
Videoconferencing	Low	High	High	High

- **Services defined in ATM**
  - Constant bit rate (telephony)
  - Real-time variable bit rate (compressed videoconferencing)
  - Non-real-time variable bit rate (video on demand)
  - Available bit rate (file transfer)

NETWORK LAYER: ROUTING

- **Routing**



A's table

initially		later	
A	–	A	–
B	B	B	B
C	C	C	C
D	B	D	B
E	C	E	B
F	C	F	B

C's table

A	A
B	A
C	–
D	D
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	–
F	F

Dest. Line

destination **next hop**

NETWORK LAYER: ROUTING

- **Routing**
  - Flooding
    - Send packet to all outgoing lines
  - Distance vector routing
    - Tanenbaum's book, p.357 – 360
  - Link state routing
    - Tanenbaum's book, p.360 – 366

- **Distance vector routing and link state routing are used in Internet**
  - We are going to review them during the discussion of IP.